

Tamanna Sharma

Profile

Dynamic and results-oriented researcher with expertise in NGS data analysis and computational strategies. Proven track record of analyzing complex biological data to provide insights and recommendations for innovative technology-based solutions. Seeking opportunities to contribute to cutting-edge research projects.

Employment History

Young Professional II (Bioinformatics), ICAR-NAARM, Hyderabad

01/2022 - 04/2023

- Collaborated with interdisciplinary teams of researchers to analyze complex problems and propose technology-driven solutions.
- NGS genomics data analysis including Whole Genome Sequencing and whole Transcriptome Sequencing.
- Compiled and interpreted data for gene expression profiling, genome annotation, and structural bioinformatics activities.
- Managed a comprehensive dashboard of the National Survey on Resources Devoted to S&T Activities (DST), ensuring accurate and up-to-date information.
- Chicken SSR Database and agricultural bioinformatics tools portal development using Django, PostgreSQL, HTML, CSS.
- Homology Modelling and Docking Studies to investigate protein-protein interactions, 3D conformations, and affinity determination.

Bioinformatician, Promea Therapeutics, Hyderabad

08/2022 - 11/2022

- Analyzed genomic sequences extracting meaningful insights for clinical and basic research applications.
- Leveraged genetic analysis techniques to explore genetic characteristics and expressions, enhancing research outcomes.

Assistant Professor, ICLES' M.J. College, Mumbai

07/2019 - 01/2021

- Designed and delivered engaging course materials for Bioinformatics and related subjects to MSc Biotechnology students.
- Led interactive classroom discussions, workshops, and practical sessions to facilitate understanding of complex concepts.
- Evaluated student performance through assignments, exams, and presentations, providing constructive feedback for improvement.
- Developed and updated curriculum to ensure alignment with the latest advancements in Bioinformatics and Biotechnology

Research Scholar, Jaypee University of Information Technology, Solan

07/2012 - 06/2018

- Molecular cloning of fatty acid and triacylglycerol biosynthetic genes in *Scenedesmus dimorphus*.
- Next-generation transcriptome sequencing of *S. dimorphus* and *S. quadricauda* and bioinformatics data analysis of the raw data.
- Differential expression analysis of genes associated with high lipid accumulation in *S. dimorphus* as compared to *S. quadricauda*.
- Computational identification of lipid-biosynthesis-related transcription factors in *S. dimorphus* and *S. quadricauda*.

Trainee Fellow, Maharshi Dayanand University, Rohtak

01/2012 - 06/2012

- Development of the Malaria Vector Database – MalVec (using HTML, PHP, and MySQL)



Personal Details

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References

Dr. Madhumohan Rao-
Katika from
ESIC Medical College and
Hospital Hyderabad

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Dr. M. Balakrishnan from
ICM Division,
ICAR-NAARM, Hyderabad

balakrishnan@naarm.org.in |

Links

<https://www.linkedin.com/in/tamanna-sharma-a6100066>

Skills

Next Generation Sequencing: WGS, WES, RNA Seq

GWAS Analysis

Data Analysis

Comparative Genomics

Gene Expression Profiling

Microsoft Office

Microsoft Excel

Programming Languages: R, Python, Perl

Web Programming: Django, HTML, CSS

Database: SQL, MySQL, PostgreSQL

Shell Scripting

Scientific Writing

Structural Bioinformatics

Education

Ph.D. Bioinformatics, Jaypee University of Information Technology,
Solan

07/2012 - 06/2018

Thesis title: Molecular Dissection of Lipid Biosynthetic Machinery in Green Algae
Scenedesmus species

M.Sc. Bioinformatics, GGDSD College , Chandigarh

07/2009 - 08/2011

B.Sc. Bioinformatics, GGDSD College, Chandigarh

07/2006 - 06/2009

Publications

- Tamanna Sharma, Rajinder S. Chauhan (2016). Comparative Transcriptomics Reveals Molecular Components Associated with Differential Lipid Accumulation between Microalgal sp., *Scenedesmus dimorphus*, and *Scenedesmus quadricauda*. *Algal Research*, 19, 109-122.
- Tamanna Sharma, Rakesh Singh Gaur, Anil Kant Thakur, Rajinder S. Chauhan (2015). Lipid content in *Scenedesmus* species correlates with multiple genes of fatty acid and triacylglycerol biosynthetic pathways. *Algal Research*, 12, 341-349.
- Rajkumar Kulandaisamy, Tushar Kushwaha, Anu Dalal, Vikas Kumar, Deepa Singh, Kamal Baswal, Pratibha Sharma, Kokkula Praneeth, Pankaj Jorwal, Srinivas Maddur, Tamanna Sharma, Sarala R Kayampeta, Manoj Kumar, Saroj Kumar, Aparoy Polamarasetty, Aekagra Singh, Deepak Sehgal, Shivajirao L Gholap, Mohan B Appaiahgari, Madhumohan R Katika, Krishna Kishore Inampudi (2022). Repurposing of FDA-approved drugs against SARS-CoV-2 Papain-Like protease by using computational, biochemical, and in vitro studies. *Frontiers Microbiology*.
- Balakrishnan M, Tamanna Sharma, P Supriya, S. K. Soam, Ch. Srinivasa Rao (2022). Next Generation Sequencing Technology: A Boon to Agriculture. *Genetic Resources of Crop Evolution*. <https://doi.org/10.1007/s10722-022-01512-5>
- Oral presentation: Homology Modeling of Photosystem I Iron-Sulfur Center (PsaC) from *Citrullus Lanatus* using Modeller. 43rd Annual Meeting of the Plant Tissue Culture Association & International Symposium on Advances in Plant Biotechnology and Nutritional Security (APBNS-2022), ICAR-National Institute for Plant Biotechnology (NIPB), New-Delhi, 28 – 30 April 2022, pp 344-345.
- Oral presentation: Homology Modeling of Capsid Protein from Mung Bean Yellow Mosaic India Virus using Modeller. 24th Annual Conference of SSCA on "Recent Advances in Statistical Theory and Applications (RASTA-2022)", ICAR - National Academy of Agricultural Research Management (ICAR - NAARM), Hyderabad, 23 – 27 February 2022.
- Poster presentation: Transcriptome Wide Computational Mining and Differential Expression of Genes Involved in Fatty Acid and Triacylglycerol Metabolism in *Scenedesmus* sp. Proceedings of the International Conference on Advances in Biomedical Engineering, Cancer Biology, Bioinformatics, and Applied Biotechnology, Jawaharlal Nehru University, New Delhi, India, 25- 26 July 2015.

Seminars and Workshops

- Training on "Applications of Bioinformatics in Agriculture" organized by ICAR-National Academy of Agricultural Research Management, Hyderabad on 15 November 2022 - 24 November 2022.
- Online training on "Prediction of Non-Coding RNA" organized by ICAR- Indian Agricultural Statistics Research Institute, New Delhi on 16 February 2022 – 18 February 2022.
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A "Bioinformatics workshop" organized jointly by the University of Nebraska, Omaha, USA, and Jaypee University of Information Technology, Solan, H.P., India from 8 May 2013-10 May 2013.

- "Computational Paradigms of Clinical Informatics" organized jointly by Louisiana Tech University, LA, USA, and Jaypee University of Information Technology, Solan, H.P., India from 1 February 2011- 3 February 2011.
- "Bioinformatics Applications in Systems Biology" in Jamia Milia Islamia University, Delhi, India from 3 March 2010 – 4 March 2010.
- National symposium on "Pharmaceutical Science and Pharmacoinformatics" organized by GGDSD College, Chandigarh, India on 10 February 2007.